Provable Learning of Linear Energy Models over Function Spaces

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1 Introduction

Abstract

Transformers with billions of parameters have achieved remarkable success across various domains, such as language modeling and computer vision. Existing theoretical works analyze the empirically observed phenomena, such as training dynamics, in-context learning (ICL) generalization, depth/width effects, scaling laws, only in isolation and in very restricted settings (single-layer, linear regression, fixed width), leaving open whether they are manifestations of a single underlying mechanism. In this paper, we show how a Transformer-like Energy-Based Model (TEBM) can provably learn a Gibbs distribution over a function space $p_\theta(f) \propto \exp(-E_\theta(f))$ from noisy observations $f_j(x_i) + \epsilon$. We consider a few examples of linear families of energy functionals and show how in the mean-field regime, where the number of heads $\to \infty$, training a TEBM recovers the parameter θ with provable approximation and generalization guarantees. Our findings prove the convergence of the unsupervised training dynamics of TEBMs, the emergence of scaling laws in the considered setting, and the necessity of the regularization of θ for the generalization properties of the ICL. We evaluate our architecture on synthetic and real-world benchmarks to support our theoretical findings.

This paper makes the following contributions:

- **Learning distributions over functions.** We formulate the problem of learning a distribution ν over a function space $\mathcal{F}$ from noisy observations as a regularized optimization problem. We model ν as an exponential family with sufficient statistic $\Phi\colon \mathcal{F} \to \mathbb{R}^d$, so that $p(f) \propto \exp(-\langle \theta, \Phi(f) \rangle)$, and state generalization guarantees for several (semi-)parametric choices of Φ.
- **Connection to in-context learning.** We draw a connection of this problem to the in-context learning in Transformer-like Energy-Based Models, discuss how it can explain the convergence to the optimal solution, the empirically observed scaling laws, the role of the positional encodings and the importance of the regularization for the generalization properties.
- **Architectural implications and empirical validation.** Guided by the optimization problem, we propose a modification of the Transformer architecture that enforces the structural assumptions of our framework. Experiments on both synthetic and real-world benchmarks demonstrate improvements in size-generalization, interpretability, and computational efficiency.

2 Related Work

Several works have showed that a fixed Transformer can implemet a gradient descent for a simple class of functions such as linear regression, Bayesian inference, etc.

3 Problem Setup

We are given a manifold $\mathcal{X}$ equipped with a probability measure μ, a function space $\mathcal{F}$ of functions $f\colon \mathcal{X} \to \mathbb{R}$, endowed with an *unknown* probability measure ν over $\mathcal{F}$. We observe a dataset $\mathcal{D} = \{D^{(1)}, \ldots, D^{(m)}\}$ consisting of m observations $D^{(j)}$ of functions $f_1, \ldots, f_m \overset{\text{i.i.d.}}{\sim} \nu(\mathcal{F})$, each evaluated at n points drawn from μ:

$$x_1, \ldots, x_n \overset{\text{i.i.d.}}{\sim} \mu(\mathcal{X}), \qquad \epsilon_{j,i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$
$$\mathcal{D} = \left\{ \{(x_i,\, f_j(x_i) + \epsilon_{j,i})\}_{i=1}^{n} \right\}_{j=1}^{m}.$$

Our goal is to learn the measure ν over $\mathcal{F}$. Given ν, predicting a new function $f \sim \nu$ from noisy observations $\{(x_i, y_i)\}_{i=1}^{n}$, where $y_i = f(x_i) + \epsilon_i$ and $\epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, reduces to MAP estimation:

$$\widehat{f} = \arg\min_{f \in \mathcal{F}} \sum_{i=1}^{n} \left| y_i - f(x_i) \right|^2 - \log \nu(f).$$

Connection to ICL in Transformers. The setup described above is closely related to in-context learning in Transformers. Given a prompt $(x_1, y_1), \ldots, (x_n, y_n)$, where x_i are positional encodings and y_i are tokens, the task is to predict the next token at position x_{n+1}. Each prompt can be viewed as a noisy observation of a latent function $f \sim \nu(\mathcal{F})$, and distinct prompts correspond to distinct draws from $\nu(\mathcal{F})$. The Transformer thus implicitly learns the shared structure across functions, that is, the measure ν itself, and uses it to generalize within a new context.

4 Linear Energy Functionals

In this section, we derive an optimization problem to address the goal described in Section 3.

Assumption 4.1. The function domain $\mathcal{X}$ is a smooth Riemannian manifold embedded in a Euclidean space $\mathcal{X} \subset \mathbb{R}^d$ equipped with a Riemannian metric $\langle \cdot, \cdot \rangle$ and a probability measure μ over it.

Assumption 4.2 (Gibbs distribution)**.** We assume that the probability measure ν over $\mathcal{F}$ is a Gibbs distribution induced by a convex energy functional $E_\theta : \mathcal{F} \to \mathbb{R}^+$ and a reference measure ([TODO: define])ν_0 on $\mathcal{F}$:

$$\nu_\theta(f) = \frac{1}{Z_\theta} \exp(-E_\theta(f))$$

where $Z_\theta = \int_{f \in \mathcal{F}} \exp(-E_\theta(f)) \nu_0(df)$ is a normalizing constant and E_θ is a family of functionals parametrized by $\theta \in \Theta$.

Assumption 4.3 (Linear energy functional)**.** We assume that the energy functional E_θ is linear in θ:

$$E_\theta(f) = \langle \theta, \Phi(f) \rangle$$

where Φ is a statistic extractor on $\mathcal{F}$.

Note, that in the general case, learning the energy functional E_θ is a hard problem because the partition function Z_θ is intractable. However, in the case of linear energy functionals for families of functions that we consider in this paper (see Section 4.1), this term is a lot easier to compute and analyze.

Lemma 4.4.

$$\nabla_\theta \log Z(\theta) = -\mathbb{E}_{f \sim \nu_\theta}[\Phi(f)]$$

Definition 4.5 (Effective dimension)**.**

$$d_{\text{eff}}(\theta) = \mathbb{E}_{f \sim \nu_\theta}[E_\theta(f)]$$

4.1 Families Of Energy Functionals

Here we look closer at specific linear families of energy functionals $E_\theta = \langle \theta, \Phi(f) \rangle$ defined by the statistic extractor Φ. We discuss what inductive bias each choice of Φ induces, what generalization properties it can achieve and how a transformer forward pass can be similar to a (sub)gradient descent for the inner problem, and transformer training can be seen as a gradient descent on the outer problem.

Table 1: Notation

Symbol	Description
$\mathcal{X}$	Domain, $\mathcal{X} \subset \mathbb{R}^d$
d	Ambient dimension
$\mathbf{S}_+^d$	Set of $d \times d$ symmetric positive semi-definite matrices
$\mathcal{Q} \subset \mathbf{S}_+^d$	Admissible set of metric tensors
$Q \in \mathcal{Q}$	PSD matrix defining an anisotropic metric
Δ_Q	Anisotropic Laplacian operator, $\Delta_Q f := \nabla \cdot (Q \nabla f)$
$\mathcal{M}(\mathcal{X})$	Space of Radon measures on $\mathcal{X}$
$\|\cdot\|_{\mathcal{M}}$	Total variation norm, $\|g\|_{\mathcal{M}} := \int_{\mathcal{X}} \lvert g(x) \rvert \, dx$

Anisotropic Sobolev regularization. Let θ be a measure over $\mathcal{Q}$. Let statistic $\Phi : H^1(M) \to \mathbb{R}^k$ be defined as $\Phi(f)(Q) := \int_{x \in \mathcal{X}} \|\nabla f(x)\|_Q^2 \, dx$. Then:

$$E_\theta(f) = \int\limits_{Q \in \mathcal{Q}} \Phi(f)(Q) d\theta(Q) \qquad \nabla_f E_\theta(f) = \int\limits_{Q \in \mathcal{Q}} \Delta_Q f d\theta(Q)$$

Anisotropic second-order TV Define statistic $\Phi : X \to \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+)$ as $\Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}}$. Then:

$$E_\theta(f) = \int_{\mathcal{Q}} \|\Delta_Q f\|_{\mathcal{M}} \, d\theta(Q) \qquad \nabla_f E_\theta(f) = \int_{\mathcal{Q}} \Delta_Q \operatorname{sign}(\Delta_Q f) d\theta(Q)$$

This promotes sparse, geometry-aligned curvature and induces a Finsler total variation.

Spectral energy features. Let $\{\psi_k\}$ be eigenfunctions of a Laplacian on M. Define $\Phi : L^2(M) \to \ell^1$, $\Phi_k(f) := |\langle f, \psi_k \rangle|^2$.

The corresponding regularizer weights energy by frequency and induces a spectral bias.

Energy type	Energy $E_\theta(f)$	Gradient / Subgradient	Hessian $\nabla_f^2 E_\theta(f)$
Anisotropic Sobolev	$\int_{\mathcal{Q}} \int_M \|\nabla f(x)\|_Q^2 \, dx \, d\theta(Q)$	$\int_{\mathcal{Q}} \Delta_Q f \, d\theta(Q)$	$\int_{\mathcal{Q}} \Delta_Q \, d\theta(Q)$ (linear operator)
Anisotropic second-order	$\int_{\mathcal{Q}} \|\Delta_Q f\|_{\mathcal{M}} \, d\theta(Q)$	$\int_{\mathcal{Q}} \Delta_Q \operatorname{sign}(\Delta_Q f) \, d\theta(Q)$ (subgradient)	Not defined (singular second variation)

Table 2: Examples of energy functionals and their variational derivatives. $\Delta_Q := \operatorname{div}(Q\nabla \cdot)$ denotes the anisotropic Laplacian.

4.2 Function Extrapolation

First, let us see how knowing the energy functional enables function extrapolation. Suppose we observe $D = \{(x_i, y_i)\}_{i=1}^n$ with $y_i = f(x_i) + \epsilon_i$, $\epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, and wish to predict $y_{m+1} = f(x_{m+1})$ at a new point x_{m+1}. Knowing the Gibbs prior $p(f) \propto \exp(-E_{\theta^*}(f)) = \exp(-\langle \theta^*, \Phi(f) \rangle)$, we recover f via MAP estimation:

$$\widehat{f}_n = \underset{f \in \mathcal{F}}{\arg\max} \; p\,(D \mid f) \; p(f) = \underset{f \in \mathcal{F}}{\operatorname{argmin}} \; \frac{1}{\sigma^2} \sum_{i=1}^n \bigl| y_i - f(x_i) \bigr|^2 + E_{\theta^*}(f).$$

This estimation will have a generalization guarantee of extrapolating f to a new point $x_{n+1} \sim \mu$ via the following theorem:

Theorem 4.6.

$$\left\| f^* - \widehat{f}_n \right\|_{L^2} \lesssim \underbrace{\|f^* - f_{\theta^*}\|_{L^2}}_{\textit{bias}} + \sigma \sqrt{\frac{d_{\text{eff}}\,(\theta^*)}{n}}$$

Correspondingly taking an expectation over $f \sim p_{\theta^*}$ yields:

$$\mathbb{E}_{f \sim p_{\theta^*}}\left[\left\|f - \widehat{f}_n\right\|_{L^2}\right] \lesssim \underbrace{\|f - f_{\theta^*}\|_{L^2}}_{\textit{bias}} + \sigma\sqrt{\frac{d_{\text{eff}}(\theta^*)}{n}}$$

For example, assuming that our function is s-smooth on $\mathcal{X} \subset \mathbb{R}^D, \dim(\mathcal{X}) = d$ then Theorem 4.6 recovers the standard rate $O\left(n^{-2s/(2s+d)}\right)$ of the minimax rate of estimation of a smooth function from n noisy observations.

5 Optimization Problem

As we mentioned, the family of the energy functionals E_θ depends on some parameter θ. Hence, our goal is to learn the parameter θ by maximizing the likelihood of the observed data. First, let us consider a simpler case, when we are given full functions $\mathcal{D} = \{f_1, \ldots, f_m\}$, not just empirical samples $f_i(x_j) + \epsilon$ and we want to learn the parameter θ by maximizing the likelihood of the observed data.

$$\max_{\theta \in \Theta} p_\theta(\mathcal{D}) = \min_{\theta \in \Theta} \frac{1}{m} \sum_{i=1}^{m} E_\theta(f_i) + \log Z(\theta)$$

where $Z(\theta) = \int_{f \in \mathcal{F}} \exp(-E_\theta(f))\nu(df)$ is a normalizing constant, ν base reference measure on $\mathcal{F}$.

Now, back to the original problem, where each function f is given by its values i n poitns x_i: $y_i = f(x_i) + \epsilon_i$, $\epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$. So given an observation $D^{(i)} = \{(x_j, y_j)\}_{j=1}^n$ our optimization problem turns into

$$\max_{\theta \in \Theta} p_\theta(\mathcal{D}) = \min_{\theta \in \Theta} -\frac{1}{m} \sum_{i=1}^{m} \log p_\theta\left(D^{(i)}\right) + \log Z(\theta) \tag{1}$$

Unfortunately, in practice each term $\log p_\theta\left(D^{(i)}\right)$ is intractable as

$$\log p_\theta(D) = \log \int_{f \in \mathcal{F}} \exp\left(-\log p(D \mid f) - E_\theta(f)\right) \nu(df)$$

5.1 Bilevel Optimization Via Masking

In this section, we discuss how to approximate our intractable convex optimization problem (1) by a bi-level non-convex optimization problem via masking approach. Fix a parameter θ and a dataset $D = \{(x_j, y_j)\}_{j=1}^n$. Define $L_\theta(f, D) := \ell(D, f) + E_\theta(f)$, where $\ell(D, f)$ denotes the empirical negative log-likelihood and E_θ is an energy functional. As the expression $-\log p(D \mid f) - E_\theta(f)$ is strongly convex, the Laplace approximation of $\log p_\theta(D \mid f)$ gives us

$$-\log p_\theta(D) \;\approx\; \underbrace{L_\theta(\widehat{f}_\theta, D)}_{\text{MAP cost}} \;+\; \underbrace{\tfrac{1}{2}\log\det H_\theta}_{\text{complexity penalty}} \tag{2}$$

where $\widehat{f}_\theta = \arg\min_f L_\theta(f, D)$ is the MAP estimator and [TODO: fix] $H_\theta = \nabla_f^2 \ell(D, \widehat{f}_\theta) + \nabla^2 E_\theta^{\text{sm}}(\widehat{f}_\theta)$.

Using this approximation, we can turn the intractable convex optimization problem (1) into a tractable bilevel non-convex surrogate of it. Given the collection of datasets $\mathcal{D} = \left\{\left(D^{(i)}, D_{\text{mask}}^{(i)}\right)\right\}_{i=1}^m$. Substituting this into Equation (1) we have

$$\max_{\theta \in \Theta} p_\theta(D) = \min_{\theta \in \Theta} \frac{1}{m} \sum_{i=1}^{m} \log p\left(D_{\text{mask}}^{(i)} \mid \widehat{f}_\theta^{(i)}\right) + E_\theta\left(\widehat{f}_\theta^{(i)}\right) + \frac{1}{2}\log\det H_\theta^{(i)} + \log Z(\theta) \quad \text{(outer problem)}$$

$$\text{where } \widehat{f}_\theta^{(i)} = \operatorname*{argmin}_{f \in \mathcal{F}} \log p\left(D^{(i)} \mid f\right) + E_\theta(f) \quad \text{(inner problem)} \tag{3}$$

The inner problem has parameter θ fixed and find the function $\widehat{f}_\theta^{(i)}$ that minimizes the loss + energy functional. The outer problem then finds the parameter θ yields the best prediction of $\widehat{f}_\theta^{(i)}$ on the masked dataset $D^{(i)}_{\text{masked}}$. Both problems are convex on their own but not jointly convex. However, we can solve it by alternating gradient descent with a following guarantee:

Theorem 5.1. *[TODO: add theorem]*

The Inner Problem. For a dataset $D = \{(x_i, y_i)\}$ we have

$$\widehat{f}_\theta = \operatorname*{argmin}_f \log p\left(D \mid f\right) + E_\theta\left(f\right) = \operatorname*{argmin}_f \|f(x_i) - y_i\|_2^2 + \langle \theta, \Phi(f) \rangle$$

Note, that due to convexity, this can be solved by subgradient descent on f with a step size $\eta > 0$:

$$f^{(t+1)} = f^{(t)} - \eta \left[2 \sum_{i=1}^{n} (f^{(t)}(x_i) - y_i)\delta_{x_i} + \nabla_f E_\theta(f^{(t)}) \right]$$

where $\nabla_f E_\theta(f^{(t)})$ is a (sub)gradient of the energy functional.

The Outer Problem.

6 Connection to Transformers And Its Implications

In most of the examples above, the key step of implementing a (sub)gradient descent for the inner problem is to compute the discrete (anisotropic) Laplacian Δ_Q for some PSD matrix Q. The original Laplacian Δ_Q can be approximated by its discrete version $\widehat{\Delta}_Q \in \mathbb{R}^{n \times n}$ computed on a set of points $x_1, \ldots, x_n \in \mathcal{X}$:

$$\widehat{\Delta}_Q = I - D^{-1/2} W D^{-1/2} \quad \text{where}$$
$$W_{ij} = \exp\left(\frac{-\langle x_i - x_j, \Pi_{x_i} Q^\dagger \Pi_{x_i}(x_i - x_j)\rangle}{\epsilon} \right) \quad \text{and} \quad D_{ii} = \sum_{j=1}^{n} W_{ij}.$$

Even though the attention mechanism is not exactly the same as the discrete Laplacian, it is similar to it in a few ways. The attention mechanism is a way to compute the weighted average ..... The discrete Laplacian is a way to compute the weighted average of the features of the points in the neighborhood of a given point. Consider a transformer-like architecture where the softmax attention is replaced with 2 different types of heads per layer:

- 1 **linear** head: $g_0(y^{(t)}) = \ldots$
- H **diffusion** heads each parametrized by some $Q_h \in \mathcal{Q}$ and $\theta_h \in \mathbb{R}^+$: $\widehat{g}_i(y^{(t)}) = \eta\theta_h \Delta_{Q_h} y^{(t)}$.

Then the forward pass is

$$y^{(t+1)} \in \mathbb{R}^n, \widehat{g}_i : \mathbb{R}^n \to \mathbb{R}^n : y^{(t+1)} = \left(I - \left[y^{(t)} - y^{(0)} \right]_{\text{mask}} - \sum_{h=1}^{H} \widehat{g}_h(y^{(t)}) \right) y^{(t)}$$

Which in the mean-field limit $n \to \infty, H \to \infty$ yields

$$f^{(t+1)} \in \mathcal{F} : g_i : \mathcal{F} \to \mathcal{F} : f^{(t+1)} = \left(I - \left[f^{(t)} - f^{(0)} \right]_{\text{mask}} - \int_{Q \in \mathcal{Q}} g_Q(f^{(t)}) d\theta(Q) \right) f^{(t)}$$

7 Experiments